Program 8

Create Kubernetes Pods using YAML Descriptors

The program demonstrates how to create and manage Kubernetes Pods using declarative YAML descriptor files instead of imperative commands.

Project Structure:

program_8/

—>nodejs-configmap.yaml

—->nodejs-pod.yaml

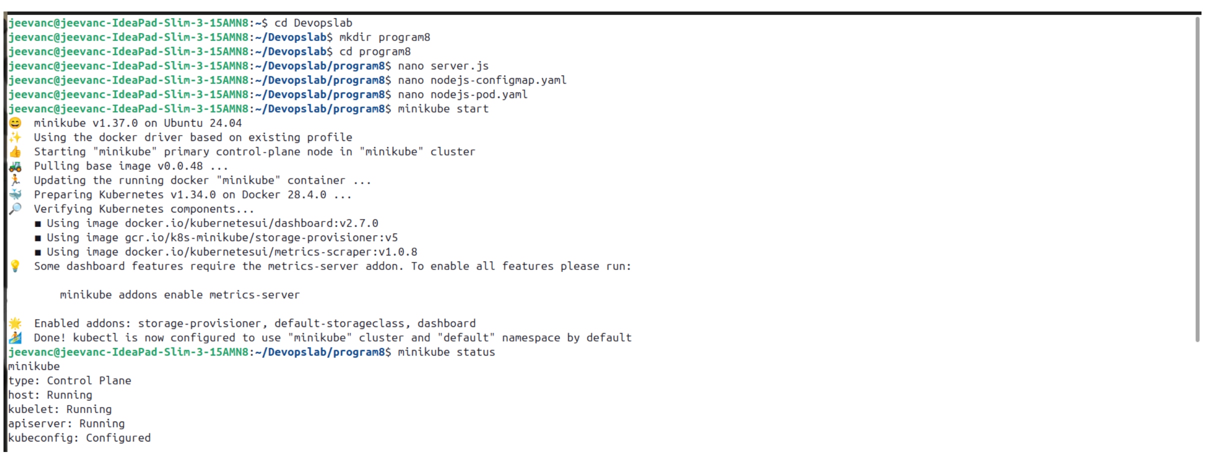
—->server.js

Procedure:

Step 1: Start Minikube and Check the status of Minikube

minikube start

minikube status



```
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~$ cd Devopslab
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab$ mkdir program8
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab$ cd program8
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ nano server.js
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ nano nodejs-configmap.yaml
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ nano nodejs-pod.yaml
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ minikube start
😄  minikube v1.37.0 on Ubuntu 24.04
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.48 ...
🏃  Updating the running docker "minikube" container ...
🐳  Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔎  Verifying Kubernetes components...
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡  Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

🌟  Enabled addons: storage-provisioner, default-storageclass, dashboard
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 2: Create nodejs-configmap.yaml file with the following code

nano nodejs-configmap.yaml

```
  GNU nano 7.2                    nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });

                              [ Read 16 lines ]
^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

Step 3: Create nodejs-pod.yaml file with the following code

nano nodejs-pod.yaml

```
  GNU nano 7.2                       nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: program-8
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config

                              [ Read 20 lines ]
^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

Step 4: Create server.js file with following code

nano server.js

```
  GNU nano 7.2                              server.js
const http = require("http");
const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from node.js running inside kubernetes");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});

                                  [ Read 10 lines ]
^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

Step 5: Apply the yaml files

kubectl apply -f nodejs-configmap.yaml

kubectl apply -f nodejs-pod.yaml

Step 6: Get the pods

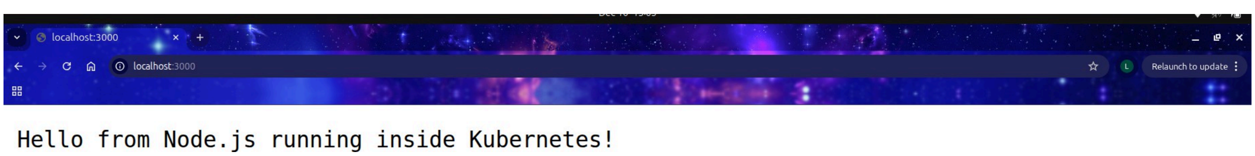
kubectl get pods

Step 7: Access the nodejs application

kubectl port-forward pod/program-8 3000:3000

Step 8: Access on the browser

http://localhost:3000



```
error: error parsing nodejs-pod.yaml: error converting YAML to JSON: yaml: line 4: found character that cannot start any token
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ nano nodejs-pod.yaml
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ kubectl apply -f nodejs-pod.yaml
error: error parsing nodejs-pod.yaml: error converting YAML to JSON: yaml: line 4: found character that cannot start any token
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ nano nodejs-pod.yaml
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ kubectl apply -f nodejs-pod.yaml
error: error when retrieving current configuration of:
Resource: "/v1, Resource=pods", GroupVersionKind: "/v1, Kind=Pod"
Name: "", Namespace: "default"
from server for: "nodejs-pod.yaml": resource name may not be empty
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ nano nodejs-pod.yaml
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ kubectl apply -f nodejs-pod.yaml
pod/program-8 created
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ kubectl get pods
NAME                                READY   STATUS              RESTARTS      AGE
kubia                               1/1     Running             2 (16m ago)   3d18h
nginx-deployment-7d684675f5-nvws7   1/1     Running             0             10m
nginx-deployment-7d684675f5-xzz8j   1/1     Running             0             10m
program-8                           0/1     ContainerCreating   0             12s
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ kubectl port-forward pod/nodejs-pod 3001:3001
Error from server (NotFound): pods "nodejs-pod" not found
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS      AGE
kubia                               1/1     Running   2 (17m ago)   3d18h
nginx-deployment-7d684675f5-nvws7   1/1     Running   0             12m
nginx-deployment-7d684675f5-xzz8j   1/1     Running   0             11m
program-8                           1/1     Running   0             84s
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ kubectl port-forward pod/nodejs-pod 3001:3001
Error from server (NotFound): pods "nodejs-pod" not found
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ kubectl port-forward pod/nodejs-pod 3000:3000
Error from server (NotFound): pods "nodejs-pod" not found
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ kubectl port-forward pod/program-8 3001:3001
Forwarding from 127.0.0.1:3001 -> 3001
Forwarding from [::1]:3001 -> 3001
Handling connection for 3001
E1212 10:39:32.988141   22786 portforward.go:424] "Unhandled Error" err=<
        an error occurred forwarding 3001 -> 3001: error forwarding port 3001 to pod cb17a5642a31bdc1a2091d956b9c4f4b8e0d8857e355497d714787bb28018209, uid : exit status 1: 2025/12/12
 05:09:32 socat[14426] E connect(5, AF=2 127.0.0.1:3001, 16): Connection refused
 >
error: lost connection to pod
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$ kubectl port-forward pod/program-8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
^Cjeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program8$
```